Getting Started with Linux Fundamentals

Course | Lab Environment

Lab: Manipulating Files and Directories

Create files by using redirection, and organize files and directories.

Modify file contents, ownership, and permissions.

Outcomes

- Organize files and directories.
- Redirect command output into files.
- Use a text editor to modify files.
- Create compressed archives.
- Use sudo privileges with commands.
- Change the ownership and permissions of files.

If you did not reset your `workstation` and `server` machines at the end of the last chapter, then save any work you want to keep from earlier exercises on those machines, and reset them now.

As the `student` user on the `workstation` machine, use the `lab` command to prepare your environment for this exercise, and to ensure that all required resources are available.

```
[student@workstation ~]$ lab start compreview-review1
```

In this exercise, you create some files on the `servera` machine. Make sure to use the exact file names from the exercise and create the files in the `/home/student` directory.

The Solution section for this lab displays command-line steps for each specification, but you can use graphical or command-line tools to perform the required tasks.

To work on this comprehensive review lab, imagine the following scenarios.

You requested a new computer and the IT Support team is ready to fulfill the request. To prepare your new workstation, the IT Support team requested some files, inventories, and back ups of the `servera` machine.

Additionally, a client asked you to organize and update some ebook files.

Specifications

- The IT Support team requested an archive called `~/user_backup.tar.gz`. This archive must contain a copy of the `/etc/passwd` file called `~/user_info.txt`, and a copy of the `/etc/shadow` file called `~/user_passwords.txt`. Remove all system accounts and leave all regular accounts in both files. If required, use sudo privileges to create the copies.

  The team also requested an inventory of the `/etc` directory. You must find all files and directories, including any items that failed the search. Save successful results in the `etc_success.txt` file, and failed results to the `etc_errors.txt` file.

  Finally, the team requested a copy of the documentation for the `mutt` email client. You can identify the documentation directory by finding a path that contains the `doc` directory. Copy all files from the resulting directory that have a `.rc` file extension to the new `~/mutt-rc` directory.

- A client requested that you organize the books in the `~/books/unsorted` directory on the `servera` machine before the books are published. Identify the category of each book by reviewing the file name. Create the necessary directories to match the following directory str and move each book to the corresponding directory:

```
books/
├── fiction
│   ├── children
│   ├── mystery
│   └── romance
└── nonfiction
│   ├── dictionary
│   ├── history
│   └── travel
└── unsorted
```

- The client also requested the following operations:

  The file name for each book includes a number; delete any file that contains a four or higher in its name.

  Change the file extension to `.pdf` for all files in the `~/books/travel` directory.

  Create the `~/books/fiction/crime` symbolic link that points to the `mystery` directory.

  Copy the `children` directory with all its contents to the `nonfiction` directory.

  Assign the `librarians` group as the owner of the `books` directory. Set all permissions for the owner user and the `librarians` group. Set read and execute permissions for others.

  Save the total size of the `books` directory in a human-readable size unit to the `books_size.txt` file. The file must look like the following example:

```
13M
```

1. As the `student` user on the `servera` machine, create the `~/user_backup.tar.gz` archive, which includes the `user_info.txt` and `user_passwords.txt` files. Copy the `/etc/passwd` file and name it `~/user_info.txt`. Remove all system account entries from the `user_info.txt` file.

   Copy the `/etc/shadow` file and name it `~/user_passwords.txt`. If required, use sudo privileges to create the copy. Make sure that the `student` user owns the `user_passwords.txt` file. In the `user_passwords.txt` file, remove all entries that do not correspond to the users that are listed in the `user_info.txt` file.

   1.1. Log in to the `servera` machine as the `student` user. Use `student` as the password.

   ```
   [student@workstation ~]$ ssh student@servera
   student@servera's password: student
   [student@servera ~]$
   ```

   1.2. Copy the `/etc/passwd` file to the `user_info.txt` file in your home directory.

   ```
   [student@servera ~]$ cp /etc/passwd user_info.txt
   ```

   1.3. Use a text editor to remove all system users. The UIDs of regular users are from 1000 to 60000.

   The `user_info.txt` file should look similar to the following example:

   ```
   [student@servera ~]$ cat user_info.txt
   student:x:1000:1000:Student User:/home/student:/bin/bash
   devops:x:1001:1001:Devops User:/home/devops:/bin/bash
   cloud-user:x:1002:1002:Cloud User:/home/cloud-user:/bin/bash
   ```

   1.4. Use sudo privileges with the `cat` command to view the `/etc/shadow` file and redirect the output to the `user_passwords.txt` file. Use `student` as the password.

   ```
   [student@servera ~]$ sudo cat /etc/shadow > user_passwords.txt
   [sudo] password for student: student
   [student@servera ~]$ ls -l user_passwords.txt
   -rw-r--r--. 1 student student 1201 Dec  3 01:17 user_passwords.txt
   ```

   1.5. Modify the `user_passwords.txt` file so that it matches all the user entries that are listed in the `user_info.txt` file.

   The `user_passwords.txt` file should look similar to the following example:

   ```
   [student@servera ~]$ cat user_passwords.txt
   student:$6$8oIjLCsc$/...:19313:0:99999:7:::
   devops:$6$MZcUmP2e$o/...:19313:0:99999:7:::
   cloud-user:!!:19340:0:99999:7:::
   ```

   1.6. Create the `user_backup.tar.gz` archive that includes the `user_info.txt` and `user_passwords.txt` files.

   ```
   [student@servera ~]$ tar --create --gzip --file user_backup.tar.gz \
   user_info.txt user_passwords.txt
   [student@servera ~]$ ls -l user_backup.tar.gz
   -rw-r--r--. 1 student student 453 Dec  3 01:24 user_backup.tar.gz
   ```

2. Create an inventory of the `/etc` directory by finding all its files and directories. Save the resulting files and directories to the `etc_success.txt` file. If the search returns any errors, then save the errors to the `etc_errors.txt` file.

   2.1. Find all files and directories in the `/etc` directory.

   ```
   [student@servera ~]$ find /etc
   /etc
   /etc/pki
   ...output omitted...
   find: '/etc/pki/rsyslog': Permission denied
   ...output omitted...
   find: '/etc/sudoers.d': Permission denied
   ...output omitted...
   ```

   The `/etc` directory contains files that the `student` user does not have permissions to view, so the command returns some error messages.

   2.2. Find all files and directories in the `/etc` directory. Redirect the results to the `etc_success.txt` file, and redirect all errors to the `etc_errors.txt` file.

   ```
   [student@servera ~]$ find /etc > etc_success.txt 2> etc_errors.txt
   ```

   2.3. Verify that the `etc_success.txt` file does not contain errors.

   ```
   [student@servera ~]$ cat etc_success.txt
   /etc
   /etc/pki
   /etc/pki/tls
   /etc/pki/tls/ct_log_list.cnf
   ...output omitted...
   ```

   2.4. Verify the contents of the `~/etc_errors.txt` file.

   ```
   [student@servera ~]$ cat etc_errors.txt
   find: '/etc/pki/rsyslog': Permission denied
   find: '/etc/dhcp': Permission denied
   ...output omitted...
   ```

3. Create a copy of the documentation for the `mutt` email client. You can identify the documentation directory by finding a path that contains the `doc` directory. Copy all files from the resulting directory that have a `.rc` file extension to the new `~/mutt-rc` directory.

   3.1. Search the entire file system for files and directories that are called `mutt`. Discard any error messages to make it easier to identify the correct location.

   ```
   [student@servera ~]$ find / -iname 'mutt' 2>/dev/null
   /usr/bin/mutt
   /usr/share/licenses/mutt
   /usr/share/bash-completion/completions/mutt
   /usr/share/doc/mutt
   ```

   The documentation directory for the `mutt` package is `/usr/share/doc/mutt`.

   3.2. Change into the directory and list the files that have a `.rc` file extension.

   ```
   [student@servera ~]$ cd /usr/share/doc/mutt
   [student@servera mutt]$ ls -l *.rc
   -rw-r--r--. 1 root root  441 May  2  2020 Mush.rc
   -rw-r--r--. 1 root root 1022 May  2  2020 Pine.rc
   -rw-r--r--. 1 root root  606 May  2  2020 Tin.rc
   -rw-r--r--. 1 root root 4833 May 11  2022 gpg.rc
   -rw-r--r--. 1 root root 1704 May 11  2022 pgp2.rc
   -rw-r--r--. 1 root root 1788 May 11  2022 pgp5.rc
   -rw-r--r--. 1 root root 1720 May 11  2022 pgp6.rc
   -rw-r--r--. 1 root root 4102 May 11  2022 smime.rc
   ```

   3.3. Create the `mutt-rc` directory in your home directory, and copy the `.rc` files from the `mutt` directory into the new directory.

   ```
   [student@servera mutt]$ mkdir ~/mutt-rc
   [student@servera mutt]$ cp *.rc ~/mutt-rc
   ```

   3.4. Verify the contents of the `mutt-rc` directory.

   ```
   [student@servera mutt]$ ls -l ~/mutt-rc
   ...output omitted...
   -rw-r--r--. 1 student student 1720 Dec  3 02:26 pgp6.rc
   -rw-r--r--. 1 student student 4102 Dec  3 02:26 smime.rc
   ```

   3.5. Return to the `student` user home directory.

```
[student@servera mutt]$ cd
[student@servera ~]$
```

4. Organize the ~/books directory contents. Review the books directory, and create the necessary directories to match the following directory structure.

```
books/
├── fiction
│   ├── children
│   ├── mystery
│   └── romance
└── nonfiction
│   ├── dictionary
│   ├── history
│   └── travel
└── unsorted
```

Each file name includes a category type that matches a directory name; move each file to the matching directory. Each file name also includes a number; delete any file that contains a four or higher in its name.

4.1. Review the books directory contents.

```
[student@servera ~]$ tree books/
books/
└── unsorted
    ├── children_book_1.epub
    ├── children_book_2.epub
    ├── children_book_3.epub
    ├── dictionary_book_1.epub
    ├── dictionary_book_2.epub
    ├── dictionary_book_3.epub
    ├── history_book_1.epub
...output omitted...
```

4.2. Delete all files that contain a four or higher in its name.

```
[student@servera ~]$ cd books/unsorted/
[student@servera unsorted]$ rm *4* *5* *6* *7*
```

4.3. Verify that the unwanted files no longer exist.

```
[student@servera unsorted]$ ls -l
...output omitted...
-rw-r--r--. 1 student student 14 Dec  2 23:35 children_book_3.epub
-rw-r--r--. 1 student student 14 Dec  2 23:35 dictionary_book_1.epub
-rw-r--r--. 1 student student 14 Dec  2 23:35 dictionary_book_2.epub
-rw-r--r--. 1 student student 14 Dec  2 23:35 dictionary_book_3.epub
...output omitted...
```

4.4. Create the required directory structure.

```
[student@servera unsorted]$ cd ~/books/
[student@servera books]$ mkdir -p fiction/mystery
[student@servera books]$ mkdir fiction/children fiction/romance
[student@servera books]$ mkdir -p nonfiction/dictionary
[student@servera books]$ mkdir nonfiction/history nonfiction/travel
```

4.5. Verify the books directory structure. Use the tree command with -d option to list only directories.

```
[student@servera books]$ tree -d ~/books
books/
├── fiction
│   ├── children
│   ├── mystery
│   └── romance
└── nonfiction
│   ├── dictionary
│   ├── history
│   └── travel
└── unsorted
```

4.6. Organize the files by moving them in the directories that match their category names.

```
[student@servera books]$ cd unsorted/
[student@servera unsorted]$ mv children* ../fiction/children/
[student@servera unsorted]$ mv dictionary* ../nonfiction/dictionary/
[student@servera unsorted]$ mv history* ../nonfiction/history/
[student@servera unsorted]$ mv mystery* ../fiction/mystery/
[student@servera unsorted]$ mv romance* ../fiction/romance/
[student@servera unsorted]$ mv travel* ../nonfiction/travel/
```

5. Manage the `books` directory and its contents per the client's request.

   Set `.pdf` as the new file extension for all files in the `~/books/travel` directory.

   Create the `~/books/fiction/crime` symbolic link that points to the `mystery` directory.

   Copy the `children` directory with all its contents to the `nonfiction` directory.

   Make sure that the `books` directory and its contents are owned by the `librarians` group. The `student` user and the `librarians` group must have permissions to read, write, and execute. Set read and execute permissions for others.

   5.1. Update all files in the `travel` directory from the `.epub` extension to the `.pdf` extension.

   ```
   [student@servera unsorted]$ cd ~/books/nonfiction/travel/
   [student@servera travel]$ mv travel_book_1.epub travel_book_1.pdf
   [student@servera travel]$ mv travel_book_2.epub travel_book_2.pdf
   [student@servera travel]$ mv travel_book_3.epub travel_book_3.pdf
   ```

   5.2. Create the `crime` symbolic link that points to the `mystery` directory.

   ```
   [student@servera travel]$ cd ../../fiction/
   [student@servera fiction]$ ln -s mystery crime
   [student@servera fiction]$ ls -l
   total 0
   drwxr-xr-x. 2 student student 90 Dec  3 02:05 children
   lrwxrwxrwx. 1 student student  7 Dec  3 02:09 crime -> mystery
   drwxr-xr-x. 2 student student 87 Dec  3 02:06 mystery
   drwxr-xr-x. 2 student student 87 Dec  3 02:06 romance
   ```

   5.3. Copy the `children` directory with all its contents to the `nonfiction` directory.

   ```
   [student@servera fiction]$ cp -r children/ ../nonfiction/
   [student@servera fiction]$ ls ../nonfiction/children
   children_book_1.epub  children_book_2.epub  children_book_3.epub
   ```

   5.4. Make sure that the entire `books` directory structure is owned by the `librarians` group. Set all permissions for the `student` user and the `librarians` groups; set read and execute permissions for others. Use your sudo privilege with the `chown` command.

   ```
   [student@servera fiction]$ cd
   [student@servera ~]$ sudo chown -R :librarians books/
   [sudo] password for student: student
   [student@servera ~]$ chmod -R 0775 books/
   [student@servera ~]$ ls -ld books
   drwxrwxr-x. 5 student librarians 56 Dec  3 02:04 books
   ```

6. Save the total size of the `books` directory in a human-readable size unit to the `books_size.txt` file.

   6.1. Check the total size of the `books` directory including all its content, and use a human-readable size unit.

   ```
   [student@servera ~]$ du -h books
   ...output omitted...
   84K     books
   ```

   6.2. Save the total size of the `books` directory to the `books_size.txt` file.

   ```
   [student@servera ~]$ echo 84K > books_size.txt
   [student@servera ~]$ cat books_size.txt
   84K
   ```

   6.3. Exit the `servera` machine.

   ```
   [student@servera ~]$ exit
   logout
   Connection to servera closed.
   [student@workstation ~]$
   ```

Hide Solution

Evaluation

As the `student` user on the `workstation` machine, use the `lab` command to grade your work. Correct any reported failures and rerun the command until successful.

```
[student@workstation ~]$ lab grade compreview-review1
```

Finish

On the `workstation` machine, change to the `student` user home directory and use the `lab` command to complete this exercise. This step is important to ensure that resources from previous exercises do not impact upcoming exercises.

```
[student@workstation ~]$ lab finish compreview-review1
```



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